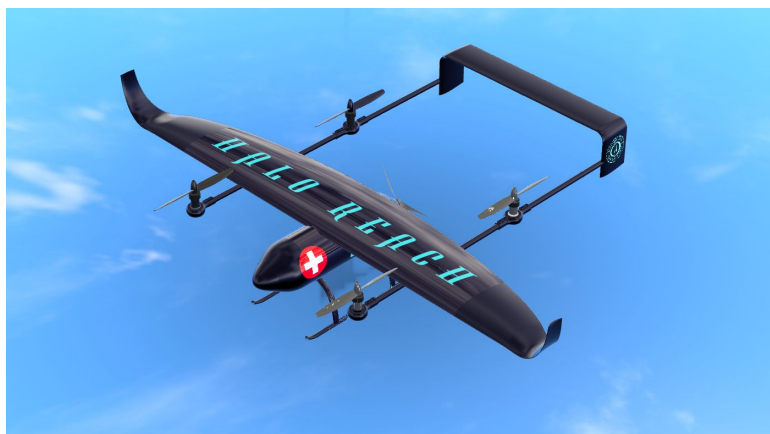

HALO REACH: *Add-on Mechanical System Design, Sizing and Calculations*

UAS for Disaster Relief - AE3 2024



Student: Adam Rosenthal

CID: 02027024

Supervisor: Dr Seifikar

Department: Department of Aeronautics

Course: MEng Aeronautical Engineering

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Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ
U.K.

Table 1. People involved in HALO REACH - GDP (**student in bold**)

Team code	Team	Names	CID
FATT01	Team Leads	Moreno, Patricia Rashid, Hasan H	01857449 02020313
FATT02	Aerodynamics	Alam, Shahriar Dirirsa, Naboth Lawlor, Max F Singh, Parasdeep Homa, Matthew M	02037395 02026211 02026199 02099678 02038403
FATT03	Structures	Leng, Jiaxun (Aaron) Wang, Zhiyan Liu, Stephen Chauhan, Shivani	02015060 01911209 02015258 02096006
FATT04	Systems and propulsion	Lin, Yongkai Chursuk, Tharm Wankhade, Divij P Aggarwal, Ishaan K Halim, George	02029838 02048996 02027906 02030969 02059571
FATT05	Flight dynamics	Baikady, Thilak S Scott-Brown, Ruairidh Allen, Harry Tang, Shun S Lee, Jeongheon (Edward)	02016794 01734080 02028116 02023280 01754474
FATT06	Add-ons	Rosenthal, Adam T Ramteke, Shreya Nakhla, Andrew	02027024 02079050 02044646

Abstract

HALO REACH is a disaster relief UAV, capable of both delivering relief supplies with a 50 kg payload, and acting as a communications relay with a 20 kg payload. The UAV is primarily designed based on the requirements of the 2024 Vertical Flight Society (VFS) Student Design Competition, but for purposes of marketability other market uses, such as for agriculture, are considered use cases for the UAV. This report details the design of the mechanism for the modular subsystem used to swap out payloads, starting from payload placement in the conceptual phase, and culminating in the choice of individual components and formation of a complete mechanism that weighs only 1.85 kg. Subsequently, some structural analysis is discussed through the selection of the rod which bears a large portion of the payload weight, which results in a design with a maximum stress of 167 MPa, giving more than a 1.5 safety factor below the 275 MPa yield strength of the aluminium rod. Finally, some discussion around ways to reduce vibrations felt by the payload is given, followed by an evaluation of the current design, including some recommendations for the next phase of the design.

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Nomenclature

σ_p	Permissible bending strength (Pa)
d	Pitch diameter (m)
F	Face width (m)
K_v	Dynamic factor
m	Gear module
N	Number of gear teeth
n	Rotational speed (rpm)
V	Pitch line velocity (m/s)
W_t	Transmitted load (N)
Y	Lewis form factor
pps	Pulses per second

1 Introduction

On the 6 February 2023, a magnitude 7.8 earthquake struck Turkey and parts of Syria. Over 50,000 people were killed and more than 107,000 were injured, with 14 million people affected overall [1]. Hatay Airport's runway suffered severe damage, which made it very difficult for relief supplies and people to be flown in. Additionally, local communication infrastructure suffered greatly [2], severely hampering rescue efforts that rely on real time communication between organisations and affected areas. Furthermore, other large-scale earthquakes have highlighted the potential for even wider communications infrastructure such as sub-sea cables to be disrupted [3]. The 2023 earthquake scenario and the risks associated with predicted increase in natural disasters motivate the need for new technology to assist in fast response to disasters, without relying on existing ground infrastructure. HALO REACH aims to fill in this gap, being able both to deliver medical supplies and to act as a communications relay to assist with disaster response immediately after a disaster occurs. The design of HALO REACH followed the brief and requirements of the 2024 Vertical Flight Society (VFS) design competition, which prescribed that the UAV be able to deliver 50kg of medical supplies after a 185 km cruise, or loiter for 10 hours above a disaster site with a 20 kg communications relay payload. The requirements of a single UAV to perform both of these different missions required a sub-system that allows different payloads to be swapped out, a 'modular add-on' subsystem. Additional relevant VFS requirements are that the payloads must be able to be swapped out within 10 minutes by two crew members, and the UAV must be VTOL capable and fit on a 6 by 6 m launch area on a ship. The existence of a subsystem allowing different payloads to be used on the UAV allows the use of HALO REACH to extend to other markets. The 10 hour loiter capability gives HALO REACH the range to be used for infrastructure monitoring, crop monitoring and general monitoring, whilst the 50 kg capacity allows the UAV to be used for crop pesticide spraying. For the overall design summary of the UAV, see the executive summary. This report focuses on initial design of the payload attachment, and the detailed selection and sizing of the mechanism. It goes on to demonstrate some structural analysis on a key load bearing member, some qualitative analysis on aeromechanical effects, and finally presents an evaluation of the design with recommendations for the next iteration of the design.

2 Conceptual Design

When choosing the high-level conceptual design for the UAV, several factors were critical to consider for the Modular Add-on subsystem team. First of all, the layout of the UAV would ideally allow for easy access to the underside of the payload, so that the requirement for the payload being swapped out by two people in ten minutes could be fulfilled. Additionally, as the disaster relief mission prescribed a 50 kg payload - just under a third of the whole UAV's mass - ideally the configuration was efficient at carrying heavy payloads. These considerations amongst others, resulted in the team preferring Vectored Thrust, VTOL Fixed Wing and Conventional Helicopter configurations. Vectored Thrust and Conventional Helicopter options would have been able to utilise the main power-producing devices to lift the aircraft, allowing for less 'dead mass' compared to the VTOL Fixed Wing option that requires VTOL Rotors capable of lifting 50 kg that would not be used throughout most of flight. Ultimately however, the required 10 hour loiter mission proved to be the most constraining for all the teams, and so it was decided to go with a conventional fixed wing to allow greatest possible efficiency for this phase, with four extra rotors to allow for vertical takeoff and landing.

2.1 Payload Placement

Initially, placing the payload on the underside of the aircraft was seen as preferable due to it's simplicity and ease of access. However, the VFS brief [4] prescribes the payload be a cube of roughly 37 cm in length. Having this large cube as opposed to a more slender rectangle below the fuselage was deemed to be infeasible due to the large amount of drag the frontal area would create. Instead, two alternative options were explored (see figure 2).

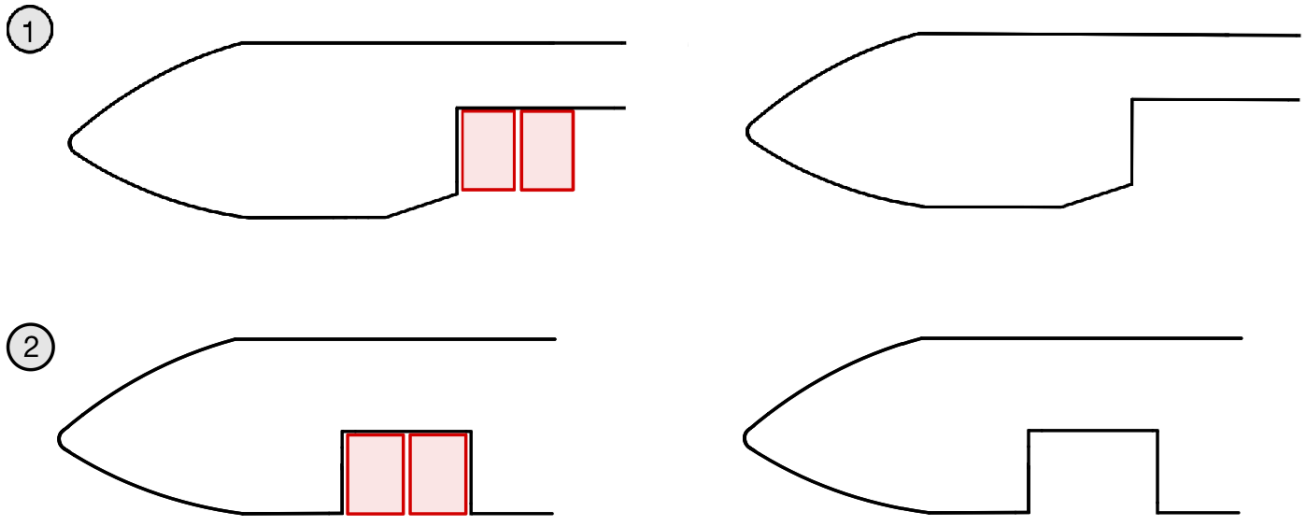


Figure 2. Payload Placement Options

Option 1 offered the advantage of easy access with the payload further from the ground, and increased options for modularity with a greater range of payload sizes able to be accommodated. On the other hand, option 2 offers more protection for the payload from the external environment, and places it closer to the centre of gravity of the aircraft - important as it represents almost a third of it's weight. Additionally, the structures team intended to place the rear connection to the landing gear in a similar position to payload option 1, which might have caused issues with the aircraft being stable on the ground. Therefore, considering potentially hazardous disaster relief environments, the better protection of option 2 resulted in the team taking it forward as the final position.

3 Mechanism Selection

After choosing the position of the payload, the primary design decision remaining was that of the attachment mechanism to stow the payload during flight, and to enable easy access, unloading and swapping when landed. Research was done into current and analogous methods of attaching payloads, culminating into a set of options to be further analysed for our use case - see table 2. These methods were analysed in the context of a Pugh matrix to assess suitability to our specific mission requirements, in order to narrow down to a few options to consider in more detail. A set of 'characteristics' were chosen as criteria to assess the options against, and then weighted based on what was required of a mechanism to fulfill the basic requirements compared to features that might have additional benefit in terms of cost, weight etc, where 10 is a very important characteristic and 1 is completely unimportant. The options were then rated based on these characteristics, with a 5 being the greatest prevalence of the characteristic, and a 1 being the least prevalence.

Table 2. Evaluation of attachment mechanism options

Characteristic	Weighting	Sliding Rails	Latch and Lock	EM Fasteners	Spring-Loaded Pins	Mechanical Clamp
Ease of use	10	4	3	5	4	2
Weight	8	3	3	2	3	3
Simplicity	7	4	4	5	5	4
Modularity	8	4	2	5	2	4
Low Power Req.	5	5	5	2	5	5
Reliability	7	5	3	3	3	5
Total		184	144	172	161	164

This allowed the choices to be ranked quantitatively, and the top 3 could be determined. As seen in table 2, sliding rails, electromagnetic fasteners and mechanical clamps were chosen as three options to assess further.

3.1 Further Analysis

In addition to initial design steps for the 3 options (see figure3, they were all assessed with some initial structural analysis [5]). There were two major issues with sliding rails and electromagnetic fasteners:

- An additional mechanism would be required to lower the payload to the ground after the mechanism was disengaged - 50 kg dropped from almost half a meter can be dangerous and damage goods.
- After delivering the payload, in the slot where the payload was housed, there would instead be a large empty space, with severe consequences regarding the aircraft's drag.

On the other hand, the inherent nature of the clamp would allow it to slowly be released, and thereby lower to the ground slowly and allow the payload to be removed. Additionally, the bottom plate of the clamp would act as a cover for the otherwise empty slot after the payload has been removed, thereby significantly removing drag penalties. The drag penalty reduction is particularly significant when interfacing with the aerodynamics team, as a very low amount of drag was deemed essential for endurance in the long 10 hour loiter phase. The main disadvantage of the mechanical clamp is that it was likely to be heavier than the other options. However, this was seen as a more manageable problem to solve than having to design additional mechanisms, as well as more robust, so the clamp was chosen as a final option - with the weight becoming a design constraint in the following design phases.



Figure 3. Three final attachment options

4 Clamp Design

With the modular attachment mechanism finalised to be a clamp, several iterations were carried out on the idea before arriving at a final design. Upon an initial weight estimation, operating with a clamping plate on the top and bottom of the payload led to a weight of close to 20 kg - far above the 7 kg allocated to the attachment mechanism by the structures team. Therefore, the team opted to use only a singular bottom plate, both to reduce weight and in order to reduce complexity with fewer parts. With one plate, the mechanism can be thought of similar to a forklift - where the payload sits on one stiff bottom plate, which can be lowered to allow easy access to unload the payload, and then raised again to sit flush with the bottom of the fuselage to reduce drag.

Next, the mechanical mechanism to operate the clamp had to be designed. The decision was made to use an electric motor to drive the mechanism as opposed to a manual lever. Although a manual lever is slightly less complex, the weight is comparable, and most importantly for the disaster relief mission; it requires much more manual effort and time to unload. As the majority of simple prime movers create rotational motion [6], the remaining design choice required the selection of a mechanism to convert rotational motion to the continuous linear motion required to lower the bottom 'clamp plate'. Some options that were considered were rack and pinion gear, lead screw mechanisms and cam and follower mechanisms. An evaluation of these three mechanisms can be see in table 3.

Table 3. Evaluation of mechanisms for converting rotational to linear motion

Mechanism	Rack and Pinion Gear	Lead Screw	Cam and Follower
Advantages	Enables smooth motion High load carrying capacity Robust and durable	High precision Self-locking capability	Able to withstand shocks Versatile Robust and durable
Disadvantages	Backlash reduces accuracy Needs lubrication	Susceptible to wear and tear Friction reduces efficiency	Can generate noise and vibration Significant space requirement Susceptible to wear and tear

Based on considerations in the table and an overall desire to reduce complexity, rack and pinion gear were chosen as a final choice. Straight teeth were chosen to avoid helical axial force component, as counteracting this would add unwanted complexity to the design.



Figure 4. Rack and Pinion mechanism

4.1 Gear selection and calculations

The choice of gear rack, pinion gear and motor themselves were primarily decided on based around a main objective of minimising mass, with secondary objectives of minimising cost and maintenance requirements. First, the rack and motor were chosen based on basic requirements for the mechanism, and subsequently the method used for choosing the pinion gear followed that recommended for the initial design phase by Childs [6].

4.1.1 Rack gear selection

The lower clamp plate is to be lowered 40 cm, which is both the ground clearance and the distance required to allow the payload to be removed from it's 'slot'. Therefore, a gear rack of at least 40 cm length was required. Furthermore, with the weight of bottom plate and payload together being less than or equal to 60 kg, the gear rack would have to be able to withstand 600 N, as a conservative

requirement in case of failure from one or all of the stringers (see figure 11). Finally, the rack was to be minimal mass. Given these constraints, the ideal choice was the rack of lightest material, at least 40 cm long that could withstand 600 N. Acetal is a plastic commonly used in gears with lighter loads [6], and is very light, and upon investigating stock suppliers was also available in a variety of sizes and tooth shapes. However, many acetal racks were not able to withstand the required load with a sufficient safety factor. Instead of switching to a stainless steel rack, however, the decision was made to incorporate two parallel rack and pinion mechanisms, which gave an overall lighter solution than heavier materials, as well as additional stability. The gear rack was chosen to have module $m = 3$, as speed is not necessary for this mechanism, and thus allows maximum torque. The rack gear was then able to be chosen from a relevant stock gear manufacturers catalogue, in this case from [7]. The important characteristics can be seen in table 4, and a full engineering drawing is included in the Appendix.

Table 4. Gear rack key information

Catalogue No.	Material	Module	Length (mm)	Bending Strength (N)	Pressure Angle	Weight (kg)
DRF3-500	Polyacetal	3	499.51	726.38	20°	0.676

4.1.2 Motor Selection

A motor selected for use with this mechanism was to be lightweight and compact whilst offering a high torque required to lift the heavy 50 kg payload, with no constraints on power requirement or rotational speed. The decision was made to use a stepper motor rather than a servo motor, due to stepper motors generally having much higher torque at low speeds and being much more reliable. Assessing the stock motors available from suppliers, many that provided torque of the order magnitude 10-20 Nm where themselves 10 kg, which was not viable due to the 7 kg restriction set on the payload attachment mechanism by the weight and balance team [8]. Therefore, a smaller torque motor with an inbuilt planetary gear configuration was chosen, in order to maintain low weight and compact size but still reaching a high enough torque (see [9]). The final choice of motor and key characteristics can be seen in table 5, and the full drawing is included in the Appendix. According to the supplier [9], the maximum torque can be provided with the motor inputting 200 pps (pulses per second). Therefore, the step angle of 1.8° means maximum torque is provided at 1 full rotation per second, or 60 rpm, which will be used in the following calculations.

Table 5. Motor key information

Catalogue No.	Max Torque (Nm)	Step Angle	Total Length (mm)	Total Weight (kg)
MST11x	11.77	1.8	90.5	0.31

4.1.3 Pinion Gear Selection

The pinion gear was chosen to be able to apply sufficient torque to lift the gear mechanism without failing. The procedure outlined in figure 5 [6] was used to select the pinion gear, which ensures that a gear choice is acceptable given a set of inputs, in this case paramount of which are the motor inputs of 11.77 Nm Torque and 60 rpm.

In order to select the number of teeth to begin the selection procedure. Each pinion gear was to lift approximately 300 N. Since the max torque available from the motor was 11.77 Nm, each pinion gear would have to lift 300 N with half the motor torque (5.885 Nm or 5885 Nmm), meaning the force would have to act over a maximum distance of 19.62 mm, thereby setting a maximum limit on the radius of the pinion gear. As per the same supplier as of the rack [7], this meant the pinion gear had to have 12 teeth, as this was the only module 3 gear with radius less than 19.62 mm. Therefore, with the number of teeth chosen, the selection verification process can begin. For small spur gears, the non-metal material available from the manufacturer was nylon, which is lightweight whilst being strong enough for the mechanism, and so looking at other manufacturers was not required. The module of the pinion was chosen to be 3 to match the gear rack, giving the highest torque possible and ensuring a constant transmission speed through perfect interaction between gears. As per figure 5, the pitch diameter $d = mN$ where m is the gear module and N is the number of teeth, was calculated to be 36 mm. The pitch line velocity was calculated by the following:

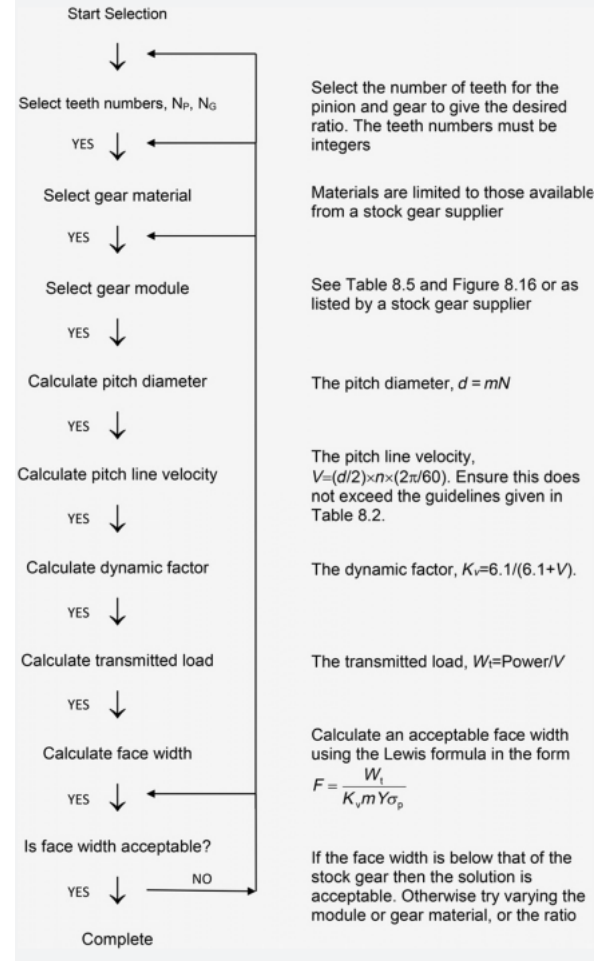


Figure 5. Gear selection procedure

$$V = \frac{d}{2} n \frac{2\pi}{60} \quad (1)$$

where V is the pitch line velocity in m/s, d is the pitch diameter in m, and n is the rotational speed in rpm, giving a value of 0.113 m/s. The dynamic factor $K_v = \frac{6.1}{6.1 + V}$ was also calculated, giving a value of 0.859. Next, the transmitted load $W_t = \frac{\text{power}}{\text{velocity}}$ was calculated, where power $P = \frac{2\pi n T}{60}$ and T is torque in Nm, yielding a transmitted load of $W_t = 300$ N. Finally, as per [6], the face width could be calculated:

$$F = \frac{W_t}{K_v m Y \sigma_p} \quad (2)$$

where F is the face width in m, Y is the Lewis form factor, and σ_p is the permissible bending stress in Pa. For this gear, the Lewis form factor was found to be 0.230 from [6]. The permissible bending stress was advised to be the material ultimate tensile strength divided by a safety factor, which in this case resulted in $\sigma_p = 24$ MPa. Finally, this allowed calculation of a face width, which was 21 mm. Following the guideline in [6]; this is less than the actual face width of 30 mm [7] and so the gear choice is acceptable. The final characteristics are shown in table 6, and a full engineering drawing can be seen in the appendix.

Table 6. Pinion gear key information

Catalogue No.	Material	Module	Pitch Diameter (mm)	Bending Strength (Nm)	Pressure Angle
PS3-12	Nylon	3	36	7.58	20°

4.2 Fasteners

The design necessitates a fastening mechanism to attach the gear rack to the bottom plate of the clamp. Careful choice of this fastener is required, as it will be responsible for carrying a large proportion of the bending load. Furthermore, the bottom plate is made out of a composite material [10], and so care has to be taken with the choice of fastener to avoid delamination of the composite layers at the interface with the fastener.



Figure 6. Steel fastener connecting rack to plate

Research [11] shows that stainless steel and titanium are the best materials for joining composites, and for HALO REACH where cost is to be minimised, steel shall be chosen as the fastening material. The specific fastener should also be of at least 50 mm in length, to allow penetration through the whole rack and at least 20 mm into the plate for a strong bond. Finally, the fastener should be easily removable for the case where the clamp need to be swapped out for other market uses, for example when a camera mount is needed at the bottom. These considerations resulted in the choice of the type of stainless screw shown in figure 6, with a square base plate.

A CAD image of the final mechanism can be seen in figure 7. The parts chosen in this section add up to a weight of only 1.85 kg for this part of the mechanism, allowing plenty of room for the design of the plate [5] and stringers [10] to allow the attachment mechanism to stay within the 7 kg limit. Note that the motor is modelled as a box to illustrate it's constraining dimensions, and the rack has been shortened to allow room for fasteners to be inserted at the bottom where the clamp will attach to the plate. A rod is used to transmit the torque from the motor to both of the pinion gears, and the following section details the design and analysis of this rod.

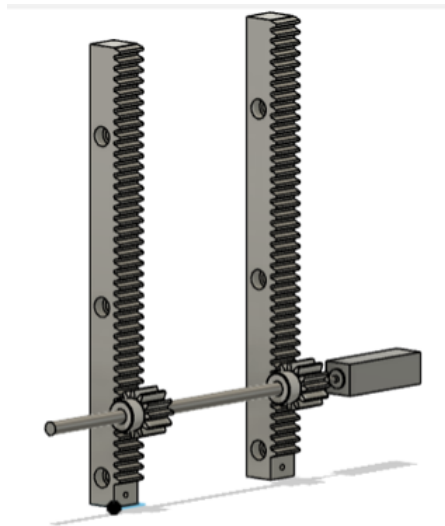


Figure 7. Final rack and pinion gear mechanism

5 Rod Analysis

5.1 Acetal Rod

After the mechanism itself was designed, the next step was to design its interaction with the larger structure of the fuselage itself. [12] details the design of heavy frames around the fuselage designed to resist the loads of the heavy components stored in the fuselage. The mechanism involves a single rod connected to the motor that transmits torque to the two pinion gears. This rod was to provide one of the ways the load from the weight of the payload was transmitted into the heavy frames, and so was to be subject to more detailed structural analysis. Initially, the rod was chosen to be polyacetal, staying consistent with the choice of plastics for the gears in order to maintain a low weight. In order to establish whether this would result in a structurally sound design, the rod was tested using Abaqus CAE software. The diameter of the rod was determined by the inner diameter of the pinion gear, in this case 12 mm. The length was set to be 400 mm to extend over the whole 369 mm payload box and to allow some room to be used to connect it to heavy frame, primarily through the use of bearings [10]. The following sections detail the analysis and results from the structural analysis simulations

5.1.1 Simulation Set-up

The rod model geometry was directly imported from Fusion 360 to be modelled as a 3D solid. The rod was modelled to be pinned at the edge faces, in order to allow for rotation required to transmit motor torque to the pinion gears. It was assumed that the rod would bear the load of the payload at two distinct points where it connects to the two pinion gears that are transmitting torque, represented as two 300 N point loads. 300 N allows some room for the final mass of the bottom plate of the clamp mechanism, as at the time of simulation this was still unknown.

5.1.2 Results

The polyacetal rod was loaded under the maximum loading conditions. The structure would have been deemed sound if the maximum stress was lower than 27.7 MPa, which results from polyacetal's yield strength of 41.55 MPa [13] with an additional 1.5 safety factor. The analyses showed that stressed in the bar rose up to 110 MPa - far above the success criteria. The difference was such that no alterations in geometry were considered - it was clear a different material was required.

5.2 Aluminium Pipe

Inspiration for further material was initially inspired by commercial pull-up bars, which tend to be of similar size as the rod in question, and rated to similar (or higher) loads. Pull up bars are typically made of steel or aluminium; since weight is a constraining factor for the attachment mechanism steel was not considered, and aluminium was pursued as the final material. Specifically, the popular 6061-T6 alloy was chosen for its all around good mechanical properties, as well as good corrosion resistance.

5.2.1 Results

Initially, to minimise mass, a pipe of inner radius of 5 mm was tested (the 6 mm outer radius remaining fixed from the gear inner diameter). Aluminium 6061-T6 has a yield strength of 275 MPa [14], and so the maximum stress criteria in this case was 183 MPa. The simulation was run under the exact same conditions as for the acetal rod, to maintain consistency. Tests showed that the maximum stress at the edges of the rod were just over 230 MPa - therefore rendering this pipe unsuitable. Therefore, the next design proposed involved a pipe with a smaller inner diameter, in this case 4 mm. This third iteration of the rod was tested under similar conditions, this time resulting in a maximum stress of 143.5 MPa. This result places it within the acceptable margin for it to be chosen as a final design. However, before completely accepting the structural integrity, some effort was made to validate this simulation result.

5.3 Simulation Validity

One major consideration in terms of solution validity originated from the method of modelling the load.

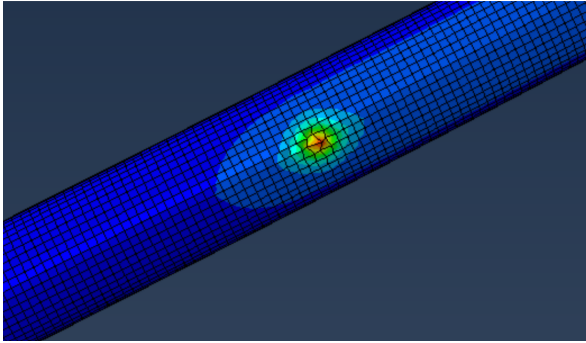


Figure 8. Stress build-up in Abaqus simulation

Exact point loads, like those used in the simulation, were not perfect representations of the actual loads on the pipe. However, they were deemed as the most suitable alternatives out of those offered by the Abaqus CAE package, and therefore an effort was made to address any discrepancies caused by loading the bar in this way. For most of the simulations, the maximum stress was found at the edges of the rods near where the rod had been fixed, which is where we would expect the highest stress to be as this area is the most constrained in deflection.

However, with certain mesh sizes, there seemed to be some unusual stress concentrations appearing at the location of the point load (see figure 8). This kind of point stress concentration in the middle of a pipe with continuous geometry is not realistic, and so was regarded as an artifact of the simulation environment and where the maximum stress was found at this kind of artefact, it was neglected - instead it was taken to be the stress at the edge of the rod.

Furthermore, a mesh convergence study was carried out on the final successful result of the Aluminium pipe with inner radius 4mm. As seen in figure 9, the simulation has converged at the finer mesh sizes, resulting in a maximum stress of 166.7 MPa. Although this is higher than the initial simulation result, it is still below the 183 MPa limit, and so this study allows full confidence in the ability of the structure to resist the loads within the allocated safety limit.

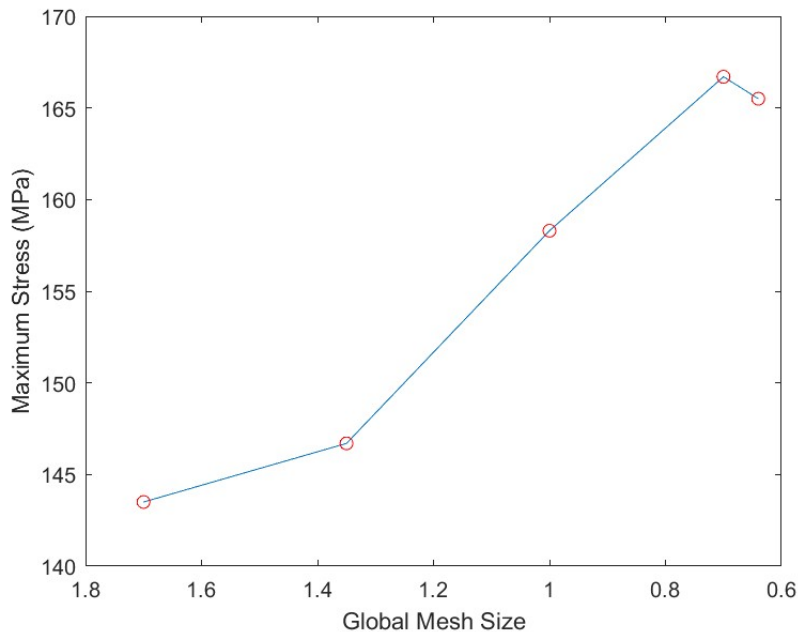


Figure 9. Variation of maximum stress with global mesh size

6 Aeromechanics and Vibrations

Broadly speaking, aeromechanics is concerned with the effect of fluid motion on bodies inside that fluid flow. One component of assessing aeromechanical effects on a UAV system involves investigation into the effect of drag caused by the payload, which are typically of large volume and are a significant source of drag. However, HALO REACH's payload was placed directly inside the fuselage to decrease drag for the longer endurance mission, and so there is much less drag from the payload. Due to the clamp plate being flat in a circular fuselage there is still some frontal area facing the flow causing additional drag (see figure 10), which has been considered further by [10]. Although drag has been mostly been accounted for in the design choices, there are other effects to consider that might affect the payload. Foremost of these is the effect on the payload of vibrations caused by both the rotating pusher propeller and the VTOL rotors. Indeed, with the recent rise of UAVs in the mainstream market, there has been significant research into handling these vibrations and their effect on payload dynamics [15]. However, much of this research concerns quadrotor UAVs with suspended payloads, where this second point in particular causing concerns with the applicability of some of these methods for HALO REACH. Instead, research into both traditional fixed-wing aircraft vibrations testing and vibrations testing for rocket launches serve as the basis for discussion of vibrations affecting HALO REACH.

6.1 Testing for vibrations

Ground Vibration Tests (GVTs) are carried out on aircraft before they receive flight certification, and are essential for testing the dynamics of the aircraft when exposed to flutter, buffeting and mechanical vibrations [16]. These tests ensure that no vibrations expected to occur in the flight envelope excite natural frequencies, causing extremely large displacements. Vibration tests are typically carried out towards the end of the design phase, where a small scale model is able to be modified with additional masses to represent the aircraft weight distribution, and thus is a low-cost method of roughly testing a range of loading frequencies. Additionally, when the final aircraft has been constructed, it will undergo a large-scale GVT to confirm that it is safe for flying. HALO REACH is only in the preliminary design phase, and thus too early in the development process for real-life model testing. Even simulations of the vibrations experienced by the payload are difficult to carry out due to the lack of detailed design in the load carrying attachments of the payload to the airframe at this current stage, as well as difficulty simulating composite materials [17] such as that of the bottom plate. Whereas weight distributions established by [8] will allow simple model testing in the following design stages, for the moment some methods of dealing with vibrations experienced by the payload are suggested to ensure it's safe arrival to deliver important disaster relief supplies.

6.2 Reducing payload vibrations

Methods of dealing with vibrations faced by payloads are extensive, particularly originating from space applications where either high-precision instruments face disruptions due to vibrations whilst in space [18] or payloads that face significant acoustic disturbances in the various phases of rocket launches. Inspiration is taken from these applications and other adjacent applications to suggest some methods to dealing with the adverse effect of vibrations for HALO REACH.

6.2.1 Negative Stiffness Isolation Systems

One class of solutions to dealing with vibrations detailed by [19] are negative stiffness isolation systems. The aim of these systems is to provide a negative stiffness to compensate for some existing elastic part of a structure, and consist of both active and passive systems. For HALO REACH, where both lightweight and simple systems are desirable for robustness in unpredictable disaster relief scenarios, only the passive systems shall be considered. The methods of introducing negative stiffness include mechanical springs, prebuckled beams, non-linear structures, magnetic structures and composite structures, and the use of these depends on resonant frequency and desired isolated frequency

band. For HALO REACH, it is likely that mechanical springs would be adopted out of these options, as they are able to isolate a large range of frequency (at least > 2.5 Hz) and are simpler and more lightweight than the others. However, more detailed consideration in the detailed design phase would be required to ensure a system that still allows for easy lowering and deployment of the payload.

6.2.2 Double wall

Another method for reducing vibrations felt by the payload is using a double walled container. Based on an application of Fahy's double-leaf partition method, which models the transmission of acoustic vibrations through a double-walled partition, [20] shows that using a payload container with an inner and outer wall, with sufficient gap between walls, the transmission to the payload can be greatly reduced. The loads experienced by the HALO REACH payload are likely to be much less than those in a launch vehicle detailed by [20], and so the proposed near-vacuum environment between walls is unlikely to be necessary. Therefore, this option remains a potentially lightweight and sufficient method of reducing vibrations felt by the payload, although will have to be considered alongside other properties necessary for the payload container such as fracture toughness, as detailed by [5].

Overall, these two methods suggest different ways of isolating the payload from vibrations in a passive way, and therefore not requiring detailed redesign of the payload and attachment mechanism. Alternative methods are also available, but until more information is required on the specific frequency band of vibrations transmitted to the payload, and the amount of decibel (dB) reduction required is available, these two methods lay the groundwork for a successful isolation system. Although isolation from vibrations has been primarily considered for the disaster relief scenario, there will be clearly benefits for other HALO REACH use cases (detailed by the executive summary), for example where a high-precision camera is needed for land monitoring.

7 Evaluation

The final design containing the full mechanism can be seen in figure 11, and the design of the other parts including the bottom plate and stringers can be seen in [10] and [5]. Overall, the design was successful in meeting the key mission constraints, and was compatible with the designs of the other sub-teams. The attachment mechanism succeeded in being simple, robust and was relatively low cost whilst fitting in the 7 kg limit imposed by the other sub-teams. However, there were a few details of the design that might be improved upon during the next iteration of the design.

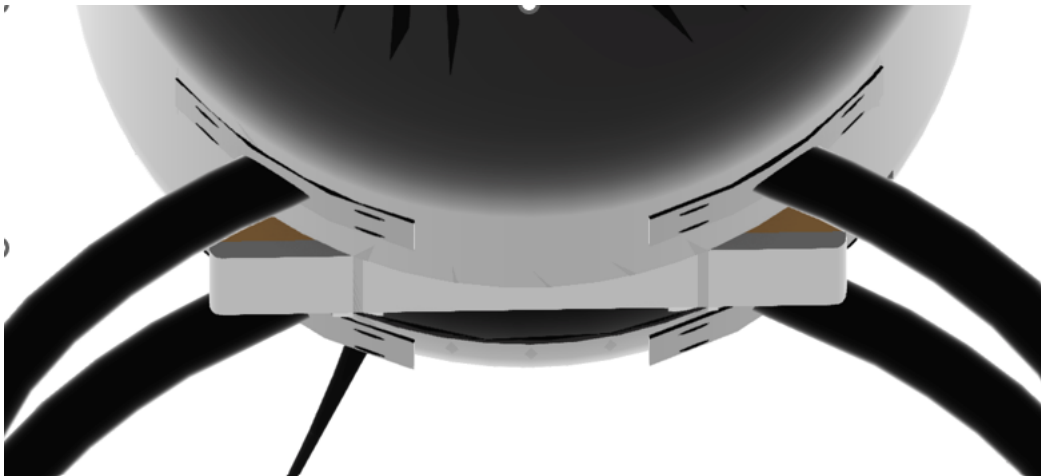


Figure 10. View of the payload extending from the payload

Firstly, upon integration of the CAD created by the modular add-on team with the final CAD from the structures team, there was clearly more of the payload 'sticking out' of the fuselage than intended

(see figure 10). Early on in the design, the initial fuselage diameter of 0.5 m had been chosen by the structures team to allow room for all the aircraft internals, whilst keeping it as small as possible, as the size of the fuselage greatly affects the drag of the aircraft. However, the add-on team required that the minimum diameter required for storing the payload would be just over 0.52 m, which is the long diagonal across the square face of the payload. Some extra clearance was added on top of this for other components in the modular mechanism, and so the final fuselage diameter was set to 0.55 m. Despite this inter-team dialogue, the clearance given seemed to not be enough, and more of the payload was sticking out of the fuselage, resulting in more frontal area and therefore a lot of additional drag. In the second iteration of the design, more internal clearance the payload and attachment mechanism was clearly required - to determine exactly how much and avoid similar underestimations, either the CAD designs should be integrated much earlier, or even a physical real size model could be created to fully understand the space constraints.

Furthermore, in the detailed design phase, more in depth analysis of the gear mechanism is needed. Although the method followed in figure 5 is suitable for this stage, it can be not considered a comprehensive structural analysis of the gears. Therefore, although specific products have been selected from catalogues to allow for initial weight and cost estimates of the attachment mechanism, these would have to be tested against more advanced analytical calculations around stress in the gears (such as those detailed in [6]) as well as simulations using FEA software.

As well as further analysis, some more design and sizing is required regarding the mechanism. Up until this point, no choices have been made regarding bearings to be placed at the motor output shaft or inside the pinion gears. These bearings would have to ensure the torque can be transferred from the motor to the rod and from the rod to the pinion gears, and must be able to withstand bending in the case of uneven loads. Therefore, choice of specific bearing, choice of material and sizing of the bearing themselves is required to fully complete the design of the clamping mechanism

Finally, as mentioned in the introduction, one requirement from the VFS brief [4] states that the payload must be able to be swapped by two crew members within 10 minutes. Whilst some rough calculations may be done to show that the mechanism is able to lower in just a few seconds, without a physical model it was impossible to prove that two crew of 'average' would be able to swap the heavy payload. That being said, although this brief point is likely to be fulfilled based on the easy payload access, an effort should be made in the detailed design phase to prove that this requirement is definitely fulfilled, perhaps through the use of a works-like prototype.

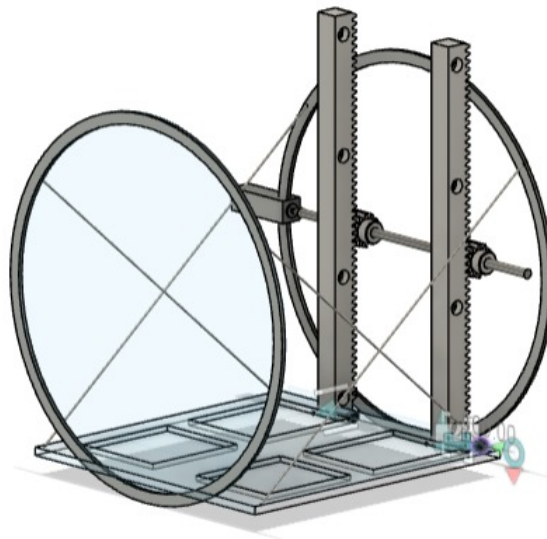


Figure 11. Final design of entire mechanism attached to two heavy frames

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Appendix

A Appendix A - Engineering Drawings

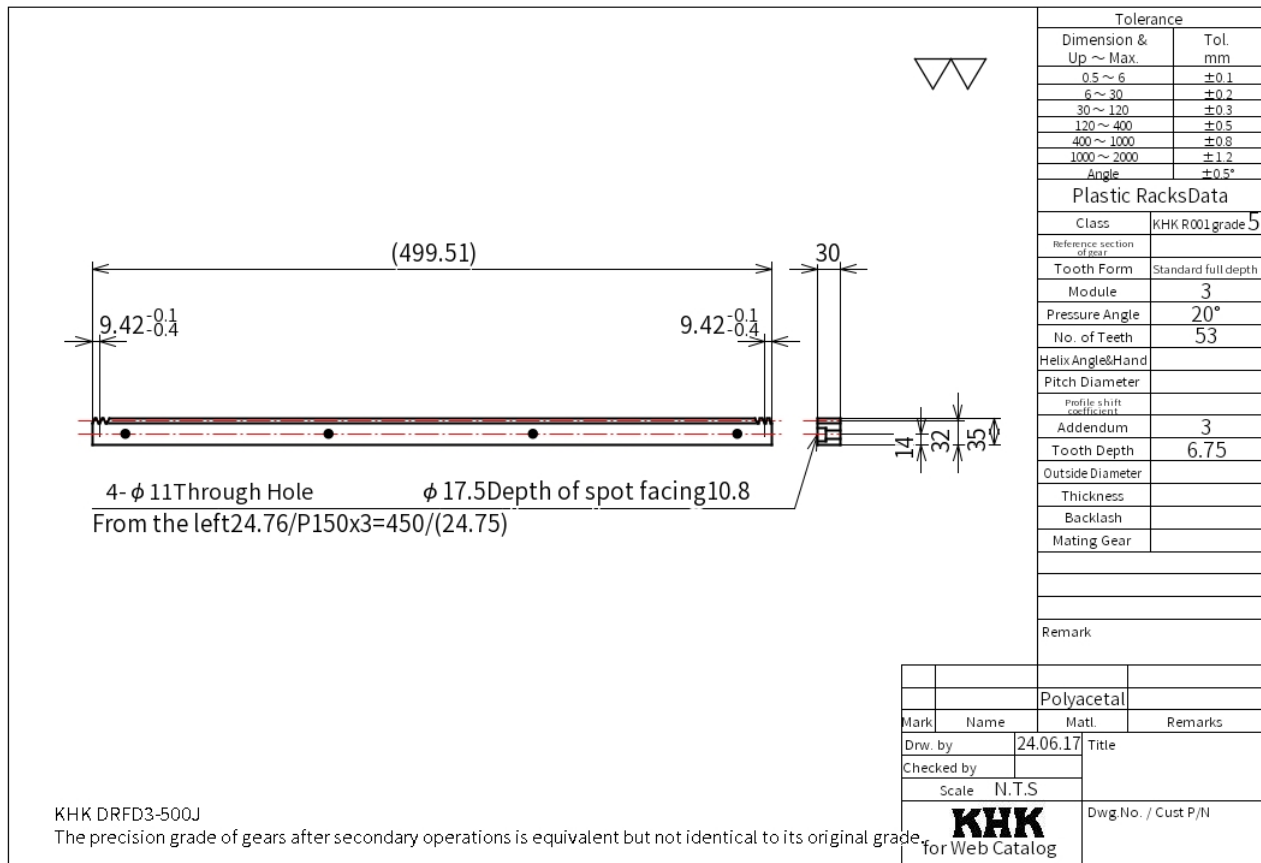


Figure 12. Rack Engineering Drawing

with double shaft only

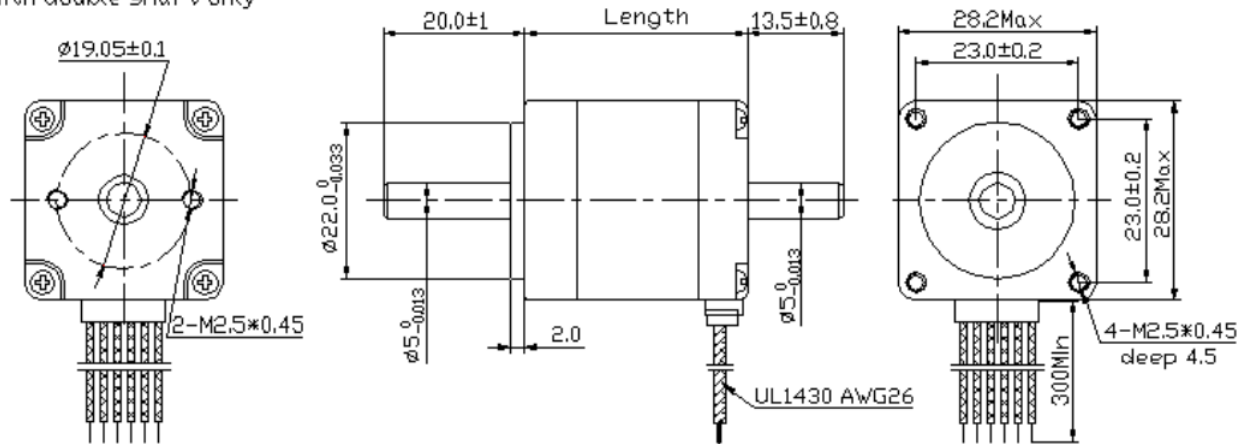


Figure 13. Motor Engineering Drawing

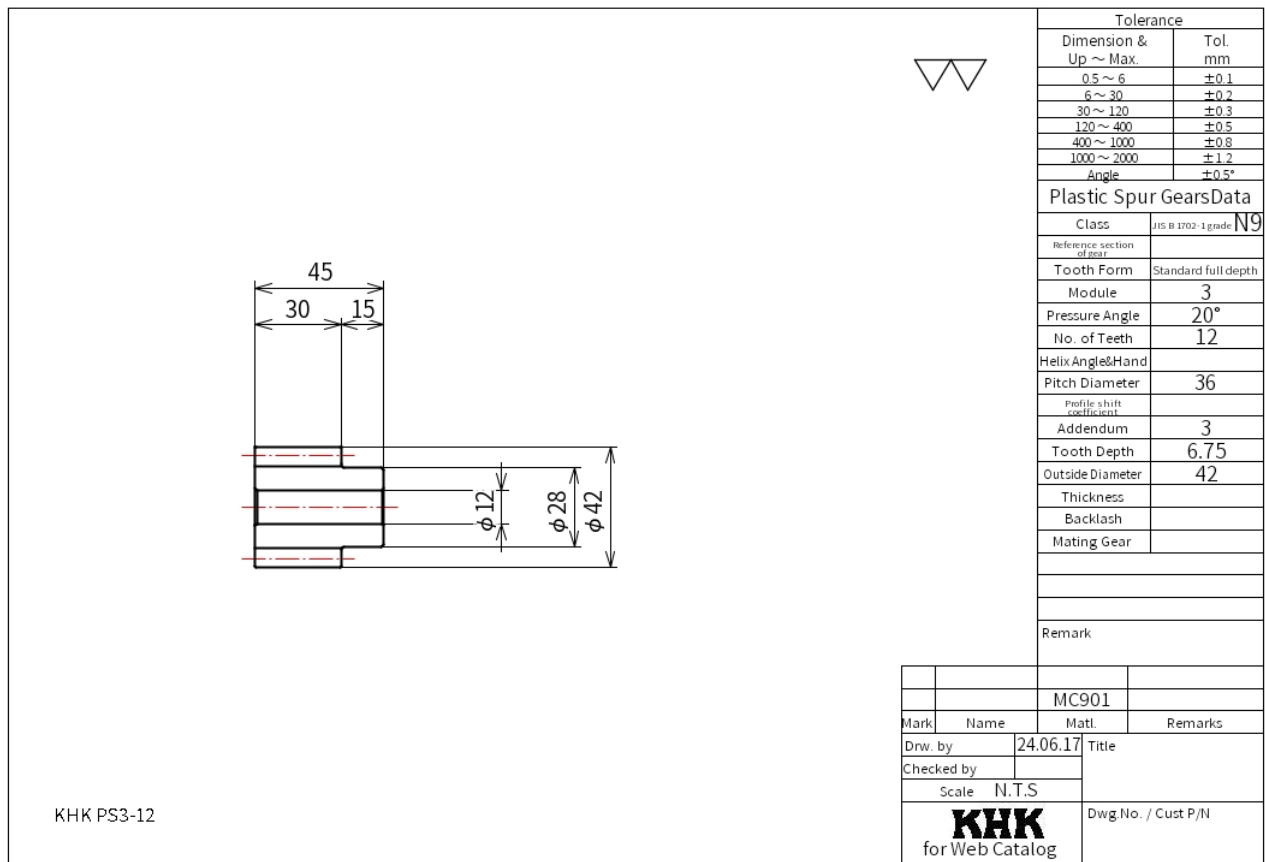


Figure 14. Gear Engineering Drawing